

```

1 function [f, iter] = abs_err_func(Re)
2 % HW_2_1 Nouh Shaikh
3
4 % Check Reynolds number bounds
5 if Re < 2500 || Re > 1e6
6     error('Re must be between 2500 and 1,000,000.');
```

7 end

8

9 % Absolute error tolerance

10 tol = 5e-6;

11

12 $g = @(f) \frac{1}{\sqrt{f}} - 4 \cdot \log_{10}(Re \cdot \sqrt{f}) + 0.4;$

13

14

15 x1 = 0.001;

16 xu = 0.1;

17

18 if g(x1)*g(xu) > 0

19 error('Choose a different initial bracket. The function does not change signs.');

20 end

21

22 iter = 0;

23 Ea = inf;

24 xr_prev = 0;

25

26

27 fprintf('Iter\t x1\t\t xu\t\t xr\t\t Ea\n');

28 fprintf('-----\n');

29

30 while Ea > tol

31 xr = (x1 + xu)/2;

32 iter = iter + 1;

33

34 if iter == 1

35 Ea = inf;

36 else

37 Ea = abs(xr - xr_prev);

38 end

39

40 fprintf('%2d\t %.6f\t %.6f\t %.6f\t ', iter, x1, xu, xr);

41 if Ea == inf

42 fprintf('%s\n', ' inf');

43 else

44 fprintf('%.8f\n', Ea);

45 end

46

47

48 if g(x1)*g(xr) < 0

49 xu = xr;

50 else

51 x1 = xr;

52 end

53

54 xr_prev = xr;

55 end

56

57 f = xr;

58 fprintf('\nFanning friction factor = %.8f\n', f);

59 fprintf('Iterations = %d\n', iter);

60 end

61